

IOT 2025–26 Final Project: Smarter Bicycle Mobility in Genoa



Urban mobility is increasingly relying on IoT technologies to improve the management, safety, and efficiency of shared and public bicycle systems. Smart bicycles equipped with GNSS receivers (and possibly other sensors e.g. PIR, cameras, etc) can collect positioning and motion data, enabling the monitoring of routes, parking behavior, usage patterns, and abnormal events. The goal of the project is to design an IoT application for the collection, transmission, analysis, and visualization of sensor data acquired from bicycles operating in the Municipality of Genoa, see (*).

In particular, the aim of the project is to design an IoT application with the following functionalities:

Monitoring of bicycle device status: Monitoring battery level, connectivity status, GNSS signal quality, sensor availability, and device health for each bicycle.

Acquisition of sensor data: Collection of real-time data generated by onboard sensors, including: GNSS position, speed, and altitude; bicycle orientation and motion data (e.g. detect anomalies monitoring the position when parked (see picture top right?); other data depending on the chosen additional sensors (IMU, PIR, cameras). Data acquisition can be performed through an embedded device, smartphone gateway, edge computer, or dedicated MCU installed on the bicycle.

Transmission of bicycle data to a central server: Communication between bicycles, edge devices, operators and cloud infrastructure using IoT protocols such as: MQTT; CoAP; WebSockets; HTTP/REST APIs; LoRa/LoRaWAN or NB-IoT for low-power wide-area communication.

Real-time and historical visualization dashboard:

Development of a dashboard for municipality operators and mobility analysts to visualize such as real-time bicycle positions on a city map; bicycle trajectories and trip history; speed and acceleration profiles; parking zones and parking violations; usage density and mobility heatmaps; availability of bicycles in different urban areas.

Definition of mobility and usage indicators:

Extraction and computation of metrics for individual bicycles and the whole fleet, such as: total distance traveled, average and maximum speed; trip duration; number of trips per bicycle; most frequent routes; parking duration and with authorized zones; detection of high-demand areas; identification of underused bicycles or areas.

Event detection and alarm generation:

Detection of critical or relevant events such as: parking outside authorized areas; bicycle entering or leaving restricted zones; abnormal motion or possible fall or incorrect parking; theft or unauthorized movement; prolonged inactivity; sensor malfunction, low battery level; dangerous riding behavior based on acceleration or vibration patterns. Notifications and alarms can be generated on dashboards or sent directly to operators through mobile applications or backend services.

Prototype requirements

In addition to a description of the proposed architecture, including motivations for the selected solutions, it is necessary to develop a prototype of the main system components, such as: sensor data acquisition module; middleware for data transmission and processing; database for storing historical sensor data; analytics module for route, parking, and event detection; dashboard for visualization and monitoring.

(*) Links:

<https://www.comune.genova.it/tutti-gli-argomenti/mobilita/bike-sharing-genova>

<https://www.voi.com/it>

<https://www.li.me/it-it/>

<https://www.bikeboom.it/>

<https://www.ridemovi.com/it/>